Supplementary Material for "Outage-Aware Reporting and Loss Recovery for Energy-Efficient River Water-Quality Telemetry over LPWANs"

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This supplement provides the details behind the main paper:

- the delivery records and the cause of their gaps (Section S1);
- the gap-length models and their tests (Section S2);
- the energy model and its assumptions (Section S3);
- further loss-handling results (Section S4);
- the reconstruction method (Section S5); and
- the protocol and further results of the design evaluation (Section S6).

References to the main paper are marked "main paper".

S1. Delivery records and the cause of gaps

Fig. S1 shows when each record was down: many short gaps, and a few gaps lasting days.

InWaterSense: The node sends each parameter as a separate message, seconds apart. The messages of one sampling instant are grouped into a reading, and a slot counts as delivered if a reading arrived for it.

Weija: Only records not marked as imputed count as delivered. Even among them, 10.4% of the conductivity values carry more than the three decimals the logger otherwise records (up to 14), a sign of unflagged interpolation. Such values hide some short gaps; therefore, counts of short gaps are lower bounds, whereas long gaps are not affected.

UVA: Missing frames are found from frame-counter gaps. A missing frame is a system loss if no sensor at all reached the database in its 10-min bin (network server, internet, or the operators' logging pipeline), a gateway loss if all of the sensor's serving gateways were silent, and a radio loss otherwise. The classification is not an artefact of the 10-min bin. With bins of 5 to 30 min, the number of system outages ranges from 14 to 8 (short ones merge), the share of system downtime in outages of a day or more from 82% to 84%, and the system share of missing frames from 28.0% to 28.6%.

LoED: The devices belong to other operators; therefore, a frame counts as lost when no LoED gateway heard it, and most devices are simply far away. Only well-heard devices are therefore used, i.e., those with at least 200 frames and a radio loss below 20% (179 of 1,780), as a proxy for a node's own link. A missing frame is a gateway loss if none of the device's serving gateways logged anything in that hour. Losses did not coincide across devices beyond what their density implies: a lost frame coincided with a loss of another device at a shared gateway within the same minute in 19.2% of cases, against 18.2% (16.7%–19.5%) under a random time-shift baseline.

Frame counters: Both LoRaWAN analyses count a frame once per device and frame counter, regardless of which gateways heard it and how often it was retransmitted. A counter decrease or a jump beyond a threshold (5,000 frames on UVA, 1,000 on LoED) is treated as a reset or rejoin, not as loss. On UVA, the frame counter decreased 1 time in the whole record, no (sensor, counter) pair was received at two different times (no counter reuse), every sensor kept its 5-min cadence in every month, and a threshold of 1,000 instead of 5,000 changes the count of missing frames by 2%. On LoED, 81 of the 179 well-heard devices show a reset, and their pooled loss is 10.0% with a threshold of 100 and 10.9% with 1,000.

Weija, the node's other streams: The node's sensors share one logger and one uplink. A conductivity slot in which another stream delivered a genuine reading (not imputed, at most three decimals, and not on an interpolated line) had a live node and a working uplink. Of the 916 missing conductivity slots, 84% were node-silent and 16% were conductivity-only gaps with the link up. In gaps of 6 h or more, only 6% of the slots had another stream delivering. The node-level record (a slot is delivered if any stream delivered) loses 13.9% of the slots in 31 gaps, with 81% of the missing slots in gaps of a day or more, close to the conductivity record's 82.9%.

InWaterSense, the second node: Both nodes reached the server through one gateway with one GPRS link, housed in node 2's manhole [1]. While node 2 was operating (May–August 2015, a window that holds 57% of node 1's missing slots), a node-1 gap during which node 2 kept delivering cannot be a gateway, GPRS, or server outage. Node 2 was silent during 79% of node 1's missing slots. The share of node-1 missing slots with a shared cause, π, follows from $P(\text{node 2 silent} \mid \text{node-1 slot missing}) = \pi + (1-\pi)\, b$, where b is node 2's own rate of silence, measured within a day around each gap at slots where node 1 delivered. Gaps of 6 h or more were approximately 74% shared-cause (gateway, backhaul, server, or a shared power failure) and 26% node-specific; gaps shorter than 6 h were 63% node-specific, as radio losses would be.

S2. Gap-length models

Besides the geometric model and the two-geometric mixture of the main paper, two heavy-tailed models are fitted to the

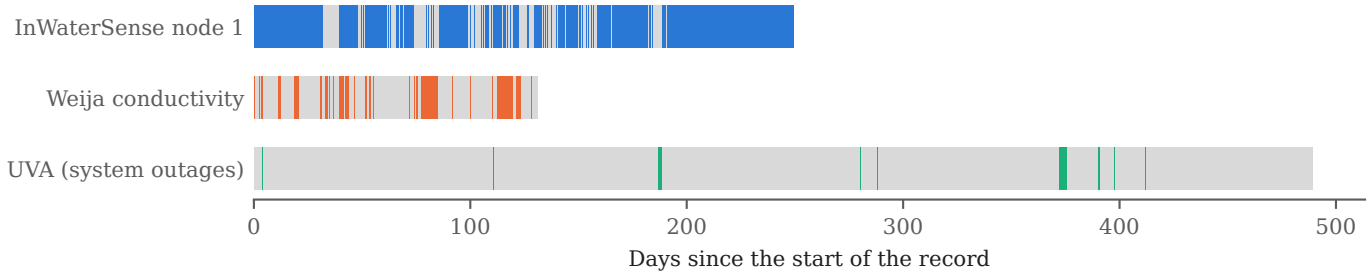


Fig. S1. Delivery timelines of the three records with long gaps: coloured spans are slots in which nothing was delivered (UVA: system outages only; its short radio losses are not shown). Each record is drawn on its own time axis, from its first slot.

gap lengths $k \geq 1$:

$$P_{\ln}(k) = \frac{\Phi\left(\frac{\ln(k+\frac{1}{2})-\mu}{\sigma}\right) - \Phi\left(\frac{\ln(k-\frac{1}{2})-\mu}{\sigma}\right)}{1 - \Phi\left(\frac{\ln\frac{1}{2}-\mu}{\sigma}\right)}, \qquad \text{(S1)}$$

$$P_{\mathrm{pl}}(k) = \frac{k^{-\alpha}}{\zeta(\alpha)}, \qquad \text{(S2)}$$

a lognormal discretised to integer lengths (S1), with Φ the standard normal distribution function, and a discrete power law (S2), with ζ the Riemann zeta function. All four are fitted by maximum likelihood and compared by Akaike's information criterion [2].

What is tested: The geometric distribution of gap lengths follows whenever the loss process has a single loss state with a constant chance of ending in each slot. This covers independent (Bernoulli) losses and the simple two-state Gilbert channel (every frame lost in the bad state, none in the good state), whose losses are correlated in time but whose loss periods still have geometric lengths. In the general Gilbert–Elliott channel both states can lose frames with probabilities below one, and loss runs are then not geometric. It is this single-loss-state assumption on gap lengths that is tested. The two-geometric mixture is a Markov chain with one up state and two loss states ("short loss" and "long gap").

Absolute fit: Because AIC only ranks models, each is also tested by a discrete Kolmogorov–Smirnov statistic $D = \max_k |\hat{F}(k) - F(k)|$ with a parametric-bootstrap p-value, using 200 samples from the fitted model, each refitted. The geometric model is rejected on every record ($p \leq 0.005$, the smallest value the 200 samples can give). With 1,619 gaps on InWaterSense and 36,519 runs on UVA, every tested model is rejected, by very different margins: the geometric model's D is 0.48 on InWaterSense and 0.26 on UVA, against 0.027 and 0.008 for the best of the others. On Weija's few gaps, the lognormal and power-law models are not rejected ($p = 0.37$ and 0.64). Without the longest gap of each record, every record keeps its best model. A gap is bounded by delivered slots on both sides; therefore, every gap length is fully observed. Only the first and last up-runs are censored, and they are not modelled.

Dependence between gaps: On InWaterSense, a long gap tends to be followed by another long gap (Spearman $\rho = +0.27$) and by a short up-run ($\rho = -0.20$; both $p < 0.001$). Gaps therefore cluster into bad periods that a renewal model, drawing each gap independently, cannot reproduce. On Weija, with 42 gaps, neither correlation is distinguishable from zero ($\rho = +0.02$, $p = 0.88$; $\rho = -0.06$, $p = 0.73$). For these reasons, all simulations replay the real records rather than a fitted model.

S3. Energy model and assumptions

NB-IoT: The measurements [3] send UDP datagrams of 12–512 B in normal coverage to a server that echoes them, without an application protocol such as CoAP. An unconfirmed report costs 0.31 J (connection released after the uplink) and a confirmed one 0.33 J (released after the server's echo, which is the acknowledgement, end to end). A bulk frame of up to 512 B, the module's largest datagram, costs up to 60% more than a 20-B report. No published measurement of the energy of a cell search when there is no network could be found. It is modelled as the receive power of 0.222 W [4] times a search time T_{search}, swept over $\{2.2, 10, 60\}$ s (default 10 s, 2.22 J per failed attempt).

LoRaWAN: Energies come from the state-by-state current profile of [5], with airtime computed from the payload. At SF7, the acknowledgement arrives in the first receive window; therefore, a confirmed success (0.164 J for a 20-B report) is cheaper than an unconfirmed report (0.244 J), which keeps both windows open. At SF12, it arrives in the second window, and a confirmed success is more expensive (0.858 vs 0.729 J). A failed confirmed attempt adds RETRANSMIT_TIMEOUT, 1–3 s [6]. Frames carry at most the regional maximum payload [6], and the 1% duty cycle forbids a new frame before 100 times the previous frame's airtime has passed.

Frames, host, and floor: A reading in a bulk frame takes 12 B (four parameters, a time offset, and a sequence number) and a frame 4 B of overhead; therefore, a frame holds 42 readings on NB-IoT, 18 at SF7, and 3 at SF12. The host is awake for every attempt, at $P_{\mathrm{on}} = 63$ mW (17 mA at 3.7 V [7]); the sensor board is off during radio attempts, which its library allows [8]. While sampling, the node draws the host's active current plus the board's constant 1.6 mA and the circuits of the operating sensors (T 3.5 mA, EC 2.5 mA, DO 0.16 mA, pH 0.17 mA [8]), $P_{\mathrm{sense}} = 92$ mW. The lifetime Y of the main paper uses the floor

$$E_{\mathrm{floor}} = 86{,}400\,(P_{\mathrm{sleep}}^{\mathrm{radio}} + P_{\mathrm{sleep}}^{\mathrm{host}}) + \frac{86{,}400}{T_{\mathrm{s}}}\,P_{\mathrm{sense}}\,T_{\mathrm{sense}}, \qquad \text{(S3)}$$

Table S1
Model inputs: measured (M) in the cited work, inferred (I) from the records, or assumed (A); where assumed, how it is tested.

Input	Value		Tested by
Link state	from the delivery records	I	gap cause, timing models
Retry timing in a gap	full slot / restoration / burst	A	all three
Reading generated in a gap	yes	A	node-off share 0–50%
Radio loss	0 or 0.096 per attempt (LoED)	I	both
Cadence	611 s, 32.5 min, 5–60 min (USGS)	I	–
Payload	20 B report; 12 B per reading + 4 B	A	–
ACK layer	LoRaWAN: network server; NB-IoT: application	M	application ACK variant
Transmissions	8 (LoRaWAN 1.0.2), NbTrans 3 or 8 (1.0.4); NB-IoT 3	M	both versions
LoRaWAN energies	state currents [5]	M	×0.5–2
NB-IoT energies	per report [3]	M	×0.5–2
NB-IoT search	0.222 W × 2.2–60 s	A	sweep
Host	63 mW awake, 0.12 mW asleep [7]	M	×0.5–2
Sensing	92 mW [8] for 5 s per sample	M, A	ratios unaffected
Storage	unlimited, free	A	RAM 2–64 KB, flash
Backlog drain	immediate	A	duty-cycle limit
Back-off cap K	6 h	A	1–24 h
Battery	6,600 mAh, ideal	A	ratios only

for sampling interval T_{s}. In (S3), the radio sleeps at 9.35 μW (NB-IoT in power-saving mode [3]) or 148 μW (LoRa radio [5]), the host sleeps at 0.12 mW (deep sleep, 33 μA [7]), and the node is assumed to be sensing for $T_{\mathrm{sense}} = 5$ s per sample. The battery holds 6,600 mAh at 3.7 V ($B = 87.9$ kJ).

Retry timing: A record shows only that the reading at a slot's first instant was lost. Retries happen at their real times, after the receive windows, the retransmission time-out, and the duty-cycle wait. Three timing models are used:

- Full slot: the link stays down for the whole slot.
- Restoration: the link returns at a uniformly random time between the last missing and the next delivered instant of each gap; therefore, only a gap's last slot can gain from a retry.
- Burst: in addition, gaps of up to 1 h are radio-loss bursts in which every retry fails with the probability of a loss following a loss on LoED (0.25), rather than surely.

Table S1 lists every model input.

S4. Loss handling: further results

ARQ: Under the full-slot model, ARQ costs 2.59×–7.36× the energy of unconfirmed reporting on LoRaWAN with eight transmissions and 1.51×–2.94× with three. On InWaterSense at SF7, this is 430.5 J/day instead of 59.3 J/day (modelled lifetime 0.46 instead of 1.63 years). With independent radio losses added, ARQ recovers them, raising the delivered share by $+5.0$ to $+7.9$ percentage points. With eight transmissions at a fixed data rate (LoRaWAN 1.0.4, NbTrans = 8), ARQ costs 6.8× the energy of unconfirmed reporting on InWaterSense at SF7, against 7.3× with the stepping of LoRaWAN 1.0.2; at SF12, there is no lower rate to step to. NB-IoT ARQ recovers nothing in any timing model because the modem reports that there is no network and the node stops retrying.

NB-IoT back-off: Searching for a cell in every slot of a gap costs 1.20×–6.54× the energy of backing off (unconfirmed reporting without back-off against unconfirmed reporting with it), and the ratio grows with the share of time in gaps and with T_{search}. With back-off, a node in a long gap searches at most once every 6 h.

Buffering: At SF7, the buffering energy relative to the fair baseline ranges from -14.9% to 0.0%, because a confirmed success costs less than an unconfirmed report and the node stops transmitting into gaps; at SF12, it is always more expensive, by $+8.0\%$ to $+25.9\%$. On InWaterSense at SF7, it uses 50.5 J/day (1.73 years). Under the other timing models, its energy stays between 0.67 and 1.08 times that of unconfirmed reporting on LoRaWAN.

Node-off gaps: When up to half of the slots in long gaps are reassigned to a node that was off, buffering still delivers every reading generated, but only 83% of the expected readings on InWaterSense and 92% on Weija; its energy relative to the fair baseline moves from 0.85 to 1.01 (SF7, InWaterSense). Weija has few long gaps; therefore, its realised node-off shares are coarse (0.49 for a nominal 0.1).

Storage and drain: The longest backlog is 22 KB. A buffer in 2 KB of RAM loses 24% of InWaterSense's readings, 8 KB loses 10%, and 64 KB none. Writing every reading to serial flash [9], with the host awake, costs at most 0.03 J/day in the datasheet's worst case, against at least 8 J/day of radio energy. If the backlog may use only what the duty cycle leaves after one live frame per slot, every reading is still delivered. At SF7, the longest backlog clears in 64 h. At SF12, the median backlog clears in 1.2 h, but in InWaterSense's worst period, a backlog persists for 63 days, and readings generated while it drains arrive up to 25 days late (95th percentile).

Back-off cap: On InWaterSense at SF7, probing in every slot costs 2.2× the energy of $K = 6$ h (3.7× on NB-IoT). Raising K from 6 to 24 h saves a further 12%, while the 90th percentile of the delay between the link's return and the arrival of the gap's last reading grows from 20 h to 61 h.

Many nodes: The discrete-event model has one EU868 gateway and N nodes that buffered through a common outage. Its rules are as follows:

- Uplinks on the same channel and spreading factor collide unless one is 6 dB stronger.
- The gateway demodulates at most eight uplinks simultaneously and cannot receive while it transmits.
- The gateway acknowledges in the first receive window if its 1% sub-band allows, otherwise in the second (869.525 MHz, SF12, 10% sub-band), and otherwise not at all.
- Nodes back off as in the main paper, drain with up to eight transmissions per frame, and respect their own duty cycle.

The collision rule reproduces the pure-ALOHA success probability e^{-2G} in a test. Table S2 gives the results.

Steady state versus outage: To separate the outage from the cell's own capacity, the same model was run for 240 h at SF12 with 25, 50, 100, and 200 nodes, after outages of 15 min (practically none) and 24 h, recording the total buffered backlog every hour. With 25 and 50 nodes, the backlog returned to a few readings per node within 72 h of either outage. With 100 nodes, the cell settled at 740–870

Table S2
N nodes recovering from a common one-day outage behind one gateway (probe jitter up to 1 h; mean of 3 runs; 72 h horizon). Cleared: nodes whose backlog was delivered; median and last clearance (h after the outage); PDR: uplinks received and acknowledged; collided; received but unacknowledged; node energy over the horizon relative to a node recovering alone.

	N	Cleared	Med. (h)	Last (h)	PDR	Coll.	No ACK	Energy
SF7	10	100%	5	6	99%	0%	1%	1.1×
SF7	50	100%	5	10	97%	1%	1%	1.5×
SF7	100	100%	5	13	93%	1%	3%	1.9×
SF7	250	100%	6	21	85%	2%	9%	3.3×
SF7	500	100%	8	42	77%	2%	15%	7.2×
SF12	10	100%	8	14	92%	2%	2%	1.3×
SF12	50	100%	15	43	67%	7%	16%	3.7×
SF12	100	93%	26	>72	47%	13%	25%	7.1×
SF12	250	36%	>72	>72	38%	18%	28%	4.7×
SF12	500	13%	>72	>72	32%	21%	29%	3.1×

buffered readings after either outage (after the 24-h outage: 9,600 readings at its end, 2,930 at 72 h, and 863 at 168 h), with approximately a third of the received frames unacknowledged. With 200 nodes, the backlog grew without bound after either outage. The capacity of a confirmed SF12 cell, not the outage, therefore sets the long-run state; the outage adds a backlog that this capacity clears.

Energy uncertainty: The ratios were recomputed 100 times with every measured component multiplied by an independent factor between 1/2 and 2 (log-uniform): unconfirmed and confirmed LoRaWAN reports, the acknowledgement time-out, the NB-IoT reports, bulk-size slope, and search power, and the host's awake power, with the NB-IoT search time drawn between 2.2 and 60 s. On LoRaWAN, ARQ cost more than unconfirmed reporting in 100% of the draws, and at least 1.8× in 97.5% of them; on NB-IoT, it cost approximately the same (median 1.01×). Hourly batching was cheaper than the fair baseline in at least 98% of the draws at SF7 and on NB-IoT, but only in 85%–91% at SF12, where it could cost up to 1.27× as much. The buffering energy relative to the fair baseline had a median of 0.81×–1.10× but ranged from 0.49× to 2.12×; its sign is set by the ratio of confirmed to unconfirmed energy.

S5. Reconstruction

All series are placed on a common 1-h grid. For every ordered pair, parameter, and gap length (1, 3, 6, 12, 24, 48, 72 h), gaps are simulated at the target (a median of 300 per pair, parameter, and length), drawn from the last 80% of the year. A gap covers the slots after the last observed slot t_0 and before the next observed slot t_1. Let y be the neighbour's series, lagged by ℓ hours, and b a scaling slope. The lag $\ell \in [0, 48]$ h maximises the correlation between 6-h changes of the target and the lagged neighbour in the first 20% of the year, and b is the least-squares slope of those changes (pairs whose best correlation is negative are dropped). Real-time estimators use only data up to t_0, and the neighbour during the gap:

$$\hat{x}_{\mathrm{pers}}(t) = x(t_0), \tag{S4}$$

$$\hat{x}_{\mathrm{diel}}(t) = x(t_0) + x(t - 24\,\mathrm{h}) - x(t_0 - 24\,\mathrm{h}), \tag{S5}$$

$$\hat{x}_{\mathrm{nb}}(t) = x(t_0) + b\,[y(t) - y(t_0)], \tag{S6}$$

$$\hat{x}_{\mathrm{dn}}(t) = \hat{x}_{\mathrm{diel}}(t) + b\,[y(t) - y(t_0)] - b\,[y(t - 24\,\mathrm{h}) - y(t_0 - 24\,\mathrm{h})]. \tag{S7}$$

Persistence (S4) holds the last value; own diel (S5) is the server's estimator of the main paper; neighbour change (S6) adds the neighbour's scaled change since the gap began; and diel + neighbour (S7) adds only the part of that change that the neighbour's own diel pattern does not explain. Bridging estimators also know $x(t_1)$:

$$\hat{x}_{\mathrm{int}}(t) = x(t_0) + \frac{t - t_0}{t_1 - t_0}\,[x(t_1) - x(t_0)], \tag{S8}$$

$$\hat{x}_{\mathrm{in}}(t) = \hat{x}_{\mathrm{int}}(t) + b\,[y(t) - \bar{y}(t)], \tag{S9}$$

where $\bar{y}$ is the straight line from $y(t_0)$ to $y(t_1)$, so that (S9) adds the neighbour's departure from its own line to the interpolation (S8). Relative gains in the main text exclude the 3 pH pairs whose target (site 14206694) has a median error of exactly zero.

Spatial structure: The neighbours lie 2.8 to 300 km from their targets (median 42 km). The order along the river is taken from the drainage area in the USGS site record, and for three sites without one, from their position on the river. If the neighbour carried water to the target, the selected lag would be positive for upstream neighbours and zero for downstream ones. Instead, for DO, a downstream neighbour had a positive lag in 64% of pairs and an upstream one in 68%.

Deep imputers: SAITS [10] and StemGNN [11] (PyPOTS implementations) see all four parameters of every site of a chain in 192-h windows, normalised per window and series from the observed values. They are trained only on the first half of 2023, with the library's random point masks and, in half of the windows, a whole site masked for 1–72 h; the checkpoint with the best error on whole-site blocks in the last 15% of the first half is kept. StemGNN uses one spectral block, one stack, and width 16 (7.5 million parameters; its layers grow with the square of the window length). A first configuration, normalised with half-year means and spreads, under-fitted by an order of magnitude even on 1-h gaps and was discarded before any comparison. The models are tested on 48,455 gap–parameter cases from the second half of the year, in which a whole site is missing for 1, 6, 24, or 72 h, and this study's estimators are scored on the same cases, with the neighbour chosen and fitted on the first half. GRIN and SPIN need a graph library that the available machine could not hold, and ImputeFormer did not finish training in the available time.

S6. Design evaluation: protocol and further results

Mechanism: Fig. S2 shows the design's behaviours in a short scenario. Before the outage, the alarm makes the event-driven node upload immediately, where plain batching would

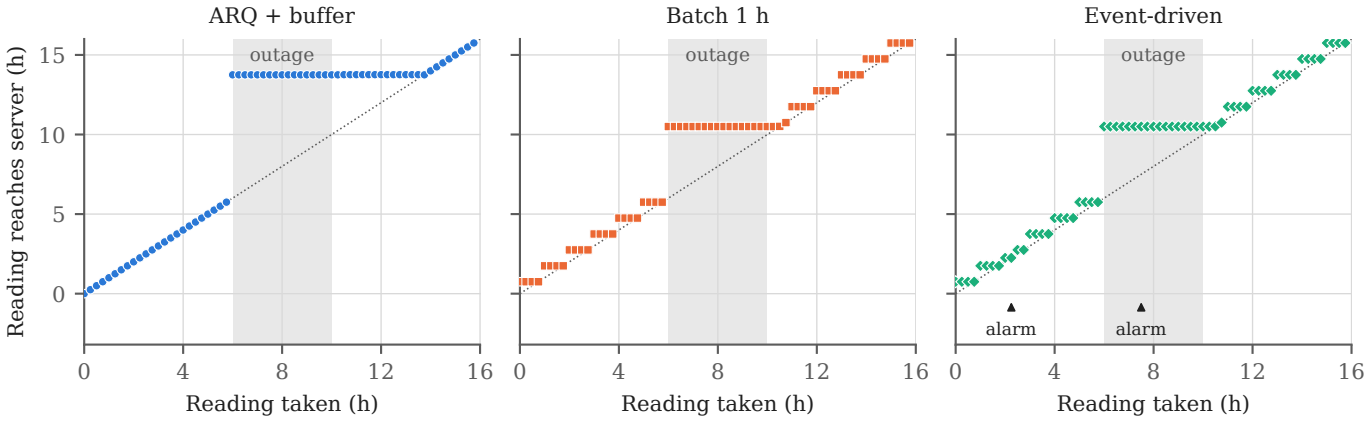


Fig. S2. When each reading reaches the server, in an illustrative scenario run through the simulator: LoRaWAN SF7, 15-min sampling, the link down from 6 to 10 h, and node alarms at 2.25 h (link up) and 7.5 h (link down). Points on the dotted diagonal arrive immediately; horizontal runs are readings delivered together. Event-driven batching follows hourly batching, except that the alarm at 2.25 h sends its batch early.

wait for the end of the hour. The alarm during the outage cannot get through. After the outage, when the backlog arrives depends on where each strategy is in its back-off.

Development phase and deviations: Hypotheses H1–H4 and their decision rules were written down before the runs for event-driven batching. After H4 failed, H5–H7 were written for the bounded-error trigger before it was implemented, and H8–H10 before they were run. Every deviation was recorded when made:

- D1: UVA replayed as system outages plus radio losses.
- D2: one site excluded because its eligibility had been computed over its record span rather than the year.
- D3: seeds derived from site identifiers.
- D4: two implementation defects of the packing order, which made H8 fail as written.
- D5 (found while the held-out runner was being built, before any 2024 data were analysed): a floating-point tie at the tolerance. pH and T are recorded in 0.1 steps, and an error of exactly one tolerance was counted in or out by rounding noise, in the scores and in the trigger.

In the first development run, H4 also failed for pH (29/48 combinations passing); those failures were caused by D5, not by batching.

Sites and links: Sampling steps are 5 to 60 min; the first 20% of each record trains the detector, and all metrics use the remaining 80%. The records are mapped onto each site's time grid in physical time, wrapped circularly with a random start offset per site (seeded by the site identifier, with a different seed in 2024). UVA is replayed as its system outages plus independent radio losses at the rate of a representative sensor (3.08% per attempt).

Injected events: For parameter j, let $a_j(t)$ be the reading minus its centred 24-h running median, and $r_j(t)$ that anomaly minus its median for the same calendar month and hour of day. The event scale is

$$\sigma_j = \max\left(1.4826\ \mathrm{MAD}[r_j]\,,\ \rho_j\right), \tag{S10}$$

floored at the recording resolution ρ_j, the most common positive difference between sorted distinct values. An event of amplitude $\kappa\sigma_j$, with σ_j from (S10), $\kappa \in \{5, 8\}$, and a random sign, lasts 180 min: a step adds $\kappa\sigma_j$ throughout, and a ramp rises linearly from 0 to $\kappa\sigma_j$; values are rounded to ρ_j. Events touch one parameter each, in turn; they lie in the test period, at least a day after the start of a data segment and wholly inside it, and at least three days apart (at most 60 per site and setting; 7,548 in 2024). Their median amplitudes are mild in physical units (5σ/8σ: T 0.74/1.19 °C, DO 7.6/12.1 %sat, EC 7.4/11.9 μS/cm, pH 0.50/0.80), and real 3-h changes of at least 5σ occur at 100% of the sites for T, 100% for DO, 93% for EC, and 60% for pH (8σ: 83%, 97%, 93%, 37%). Of the 2024 events, 99.7% stay within T −0.5–40 °C, positive EC, pH 4–11, and DO 0–250 %sat (99.8% in 2023), and the latency verdicts also hold on the plausible events alone.

Detector: The detector compares each reading with the reading $m = 120$ min earlier, corrected by a learned hour-of-day profile $p(\cdot)$ of the departure from the running 24-h median, and runs a two-sided CUSUM [12] on the clipped, standardised residual:

$$z_t = \mathrm{clip}\left(\frac{x_t - x_{t-m} - p(h_t) + p(h_{t-m})}{s},\ -c,\ c\right), \tag{S11}$$

$$S_t^+ = \max\left(0,\ S_{t-1}^+ + z_t - k\right), \quad S_t^- = \max\left(0,\ S_{t-1}^- - z_t - k\right), \tag{S12}$$

where h_t is the hour of day and s is a robust residual scale floored at ρ_j. An alarm is raised when either sum in (S12), driven by (S11), exceeds H, with $k = 3$, $H = 8$, and $c = 6$; the detector then stays in an event state until $|z_t| < 2$ for 60 min. The profile and scale are initialised on the training period and updated by exponential smoothing (memory of approximately 7 days) outside the event state. These parameters were fixed before any evaluation. In 2024, the detector caught a median (over sites) of 67% of 5σ steps, 91% of 8σ steps, 13% of 5σ ramps, and 65% of 8σ ramps, with a median of 0.8 false alarms per parameter per 30 days (mean 1.1).

All injected events: Counting every injected event, detected or not, the server knew 34% within an hour with event-driven batching on the no-outage link at SF7, the same as with per-reading reporting, against 21% with hourly batching; within a

Table S3
Held-out year (2024): share of scored slots within the L2 tolerance (median over sites; LoRaWAN SF7; series without injected events). ARQ: ARQ + buffer.

Link	Param.	Unconf.	ARQ	Batch	Event	Bounded
No outage	DO	100.0%	100.0%	91.8%	91.8%	100.0%
	pH	100.0%	100.0%	99.0%	99.1%	100.0%
	T	100.0%	100.0%	98.7%	98.7%	100.0%
	EC	100.0%	100.0%	96.7%	96.7%	100.0%
UVA	DO	98.8%	99.2%	91.3%	91.5%	99.2%
	pH	99.8%	99.8%	98.5%	98.6%	99.8%
	T	98.9%	98.8%	97.7%	97.7%	98.8%
	EC	99.2%	99.3%	96.5%	96.6%	99.3%
Weija	DO	90.4%	89.6%	83.8%	83.8%	89.4%
	pH	95.3%	95.0%	94.1%	94.1%	95.4%
	T	89.0%	88.2%	86.9%	87.0%	88.2%
	EC	90.5%	89.6%	87.2%	87.2%	89.7%
InWaterSense	DO	75.7%	72.4%	67.7%	67.7%	72.4%
	pH	90.9%	88.3%	90.1%	90.2%	90.6%
	T	74.1%	67.7%	67.2%	67.5%	68.2%
	EC	80.5%	74.6%	74.0%	73.8%	75.1%

Table S4
Held-out year (2024): detected events whose onset fell in a gap (LoRaWAN SF7, pooled over sites): number of events, share the server ever learns, and median latency of those it learns (hours).

		Known ever			Median latency (h)			
Link	n	Unconf.	ARQ	Event	ARQ	Batch	Event	Bounded
InWaterSense	2,113	3.0%	98.8%	98.8%	27.5	26.6	24.3	24.0
Weija	661	4.1%	100.0%	100.0%	53.5	48.5	48.5	48.5
UVA	57	7.0%	100.0%	100.0%	28.5	24.9	24.9	24.9

day, it knew 58% with any of them. On the InWaterSense link, the shares within an hour were 19% (bounded), 16% (ARQ + buffer), 12% (unconfirmed), and 11% (batch), and within a day, 44%, 42%, 22%, and 43%.

Picture: Table S3 gives the picture for every parameter. Under the two water-quality gap records, unconfirmed reporting showed a better picture than ARQ + buffer for every parameter and radio (DO on InWaterSense at SF7: 75.7% vs 72.4%). When the link returns, an unconfirmed node delivers its next reading immediately, while a buffering node waits for its next back-off probe.

Events that start during a gap: Table S4 and Fig. S3 show the detected events whose onset fell in a gap. Unconfirmed reporting learned almost none of them (at most 7.0% on any link and radio); the buffering strategies learned at least 98.8% after the gap.

Packing order (H8): On SF12, a frame holds only 3 readings; therefore, an hourly batch at 15-min sampling needs two frames, and the duty cycle delays the second. With first-in-first-out packing, the newest reading rode in the second frame, and in the first development run, the picture on the no-outage link fell to 98.72% at worst. As written, H8 failed at one site (01467200, 5-min sampling, SF12; minimum 99.998%). The cause was two implementation defects. First, the reading promoted to the first frame could be a placeholder for a sensor gap. Second, the node's copy of the server counted readings as known as soon as their upload began, although at 5-min sampling an SF12 upload spans several slots. With both fixed (D4), newest-first packing was within tolerance at every scored slot of the no-outage link, on every site, parameter, and radio, in both years, at 0.987–1.007 times the energy of first-in-first-out packing.

Table S5
Held-out year (2024): energy (median J/day over sites, LoRaWAN SF7, series without injected events) at three tolerance levels, the median over sites of each site's ARQ / Bounded energy ratio, and the lowest within-tolerance share over the four parameters (median site). ARQ: ARQ + buffer.

		Energy (J/day)			Within tolerance	
Level	Link	ARQ	Bounded	ARQ/Bounded	Bounded	ARQ
L1 (tight)	No outage	32.6	19.1	1.56×	100.0%	100.0%
	InWaterSense	35.6	25.5	1.27×	61.9%	62.5%
L2	No outage	32.6	10.9	2.53×	100.0%	100.0%
	InWaterSense	35.6	20.3	1.65×	68.2%	67.7%
L3 (loose)	No outage	32.6	9.0	3.39×	100.0%	100.0%
	InWaterSense	35.6	18.5	1.83×	79.1%	78.3%

Tolerance level (H9, H10): Table S5 provides the energies at the three tolerance levels.

Tolerances and sensor accuracy: L2 lies at or inside the U.S. Geological Survey's rating of an "excellent" continuous record [13]. Over the sites used, the median L2 tolerance is 0.20 °C against 0.20 °C for T, 2.0 against 4.8 %sat for DO (0.3 mg/L or 5% of the reading), 4.2 against 6.3 μS/cm for EC (3% of the reading; USGS rates conductance in percent, as the EC tolerance is defined), and 0.10 against 0.20 for pH. All three levels are small compared with the median daily range at these sites (T 1.10 °C, DO 10.1 %sat, EC 12.0 μS/cm, pH 0.25).

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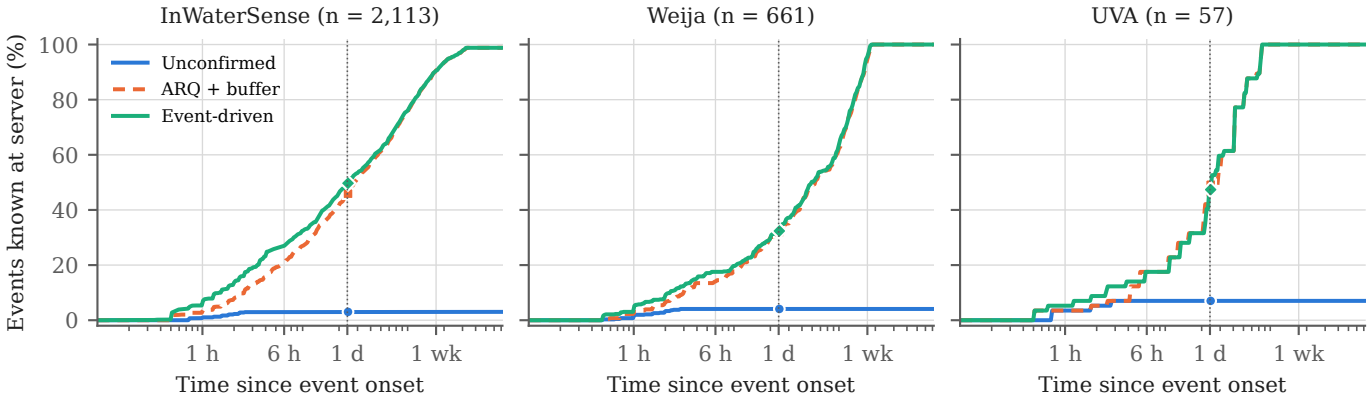


Fig. S3. Held-out year (2024): detected events whose onset fell in a gap, share known at the server by the time on the horizontal axis (LoRaWAN SF7, pooled over sites; markers at 1 day).

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